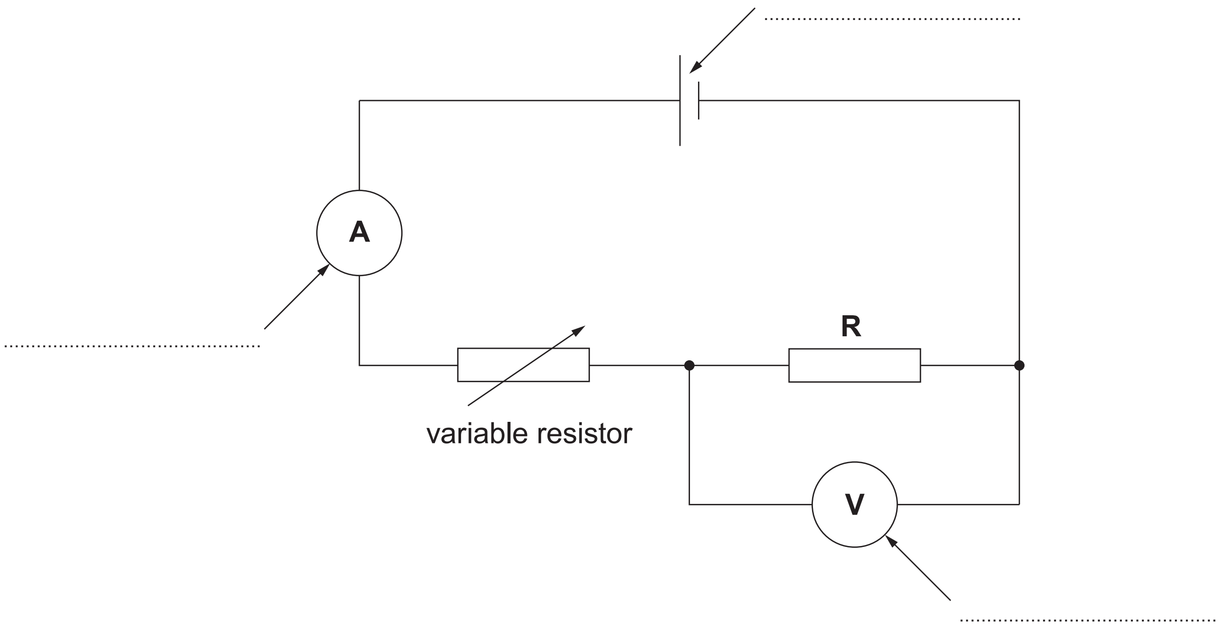
4. Some pupils set up the circuit below to measure the current in a resistor (**R**) and the voltage across it. The circuit is used to give a number of readings.



(a) (i) **Label** the parts of the circuit diagram shown above. [3]

(ii) State the purpose of the variable resistor. [1]

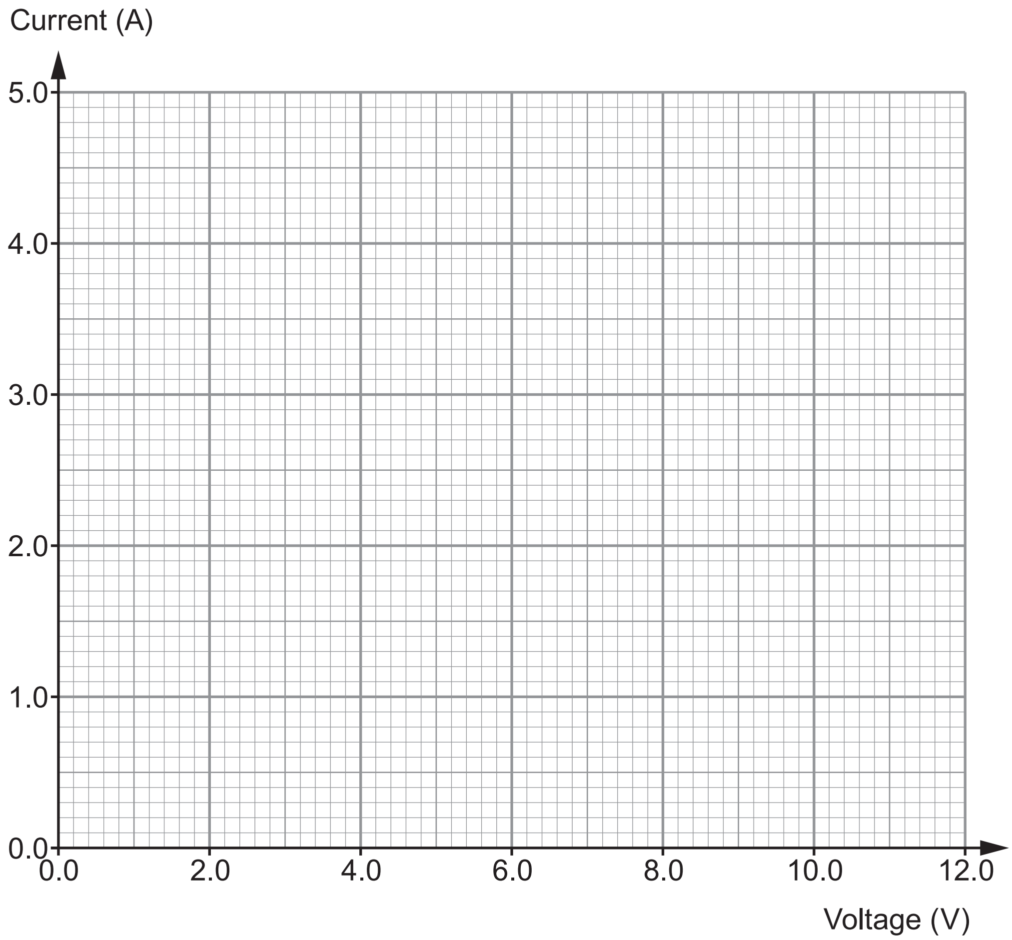
....

Examiner only

(b) The circuit was used to obtain the following results.

Voltage (V)	Current (A)
0.0	0.0
2.0	0.8
4.0	1.6
6.0	2.4
10.0	4.0
12.0	4.8

(i) Plot the data on the grid below and draw a suitable line. [3]



Examiner only

(ii) Use your graph and the equation:

$$\text{resistance} = \frac{\text{voltage}}{\text{current}}$$

to calculate the resistance of the resistor at 8.0 V. [2]

Resistance = ... Ω

(iii) Complete the following sentence about the trend on the graph by underlining the correct word or phrase in each bracket. [3]

As the voltage increases, the current (**decreases** / **stays the same** / **increases**)

at a (**decreasing** / **constant** / **increasing**) rate. As a result, the resistance

(**decreases** / **stays constant** / **increases**).

(iv) One pupil in the group, Billy, made the statement that the power of the resistor increases as the voltage increases. By using an equation from page 2, explain whether you agree with Billy. [2]

....

....

....

(v) A second group of pupils use a resistor that has **half** the resistance of the one in this experiment. **Draw a line on the original grid** to show how the current through it would change with voltage. [2]

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